UNC Charlotte

ECGR 3120 Electromagnetics

Magnetically Steered Cardiac Ablation

Milestone 1: Navigation Magnetic Field

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Submission date: _____

Input status: The 80 mT center-field case in this report is illustrative. It has not been verified against the professor's official starter kit or baseline values.

Abstract

This report documents a numerical magnetic-field model for two coaxial Helmholtz coils. A vectorized, midpoint-discretized Biot–Savart calculation is compared with analytical circular-coil and Helmholtz-pair fields. The reported results include a segment-convergence study and field-uniformity estimates over the central working region. The 80 mT center field is an illustrative example used to set the ampere-turns; it has not been verified against the professor's official starter kit or baseline values.

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1 Objective

The objective of Milestone 1 is to calculate the magnetic flux density **B** produced by a pair of Helmholtz coils and assess the accuracy of a numerical Biot–Savart calculation against analytical electromagnetic theory. The project handout describes a central working region extending $\pm 50\,\mathrm{mm}$ about the coil-pair center. A useful navigation field should be relatively uniform within this region.

2 Geometry, inputs, and assumptions

The modeled coils are circular, coaxial loops centered on the z-axis. Each coil has radius

$$a = 0.35\,\mathrm{m},$$

and the distance between coil centers is

$$d = 0.35\,\mathrm{m} = a.$$

Thus, the coil planes are at $z = -0.175\,\mathrm{m}$ and $z = +0.175\,\mathrm{m}$. The working-region half-width is $0.05\,\mathrm{m}$.

2.1 Illustrative field input

No official coil current or turn count was found in the provided project files. The standalone script therefore assumes an *illustrative* Helmholtz center field of $80\,\mathrm{mT}$. This choice is used only to infer the ampere-turns NI for each coil from the analytical center field equation. The resulting value is

$$NI = 31\,139.611\,275\ \mathrm{A}\cdot\mathrm{turn}.$$

This product does not specify a separate current and turn count, and it is not an official professor-provided design value. All results scaled by NI, including the field strengths in the plots, should be treated as an illustrative case until the official starter kit or values are supplied.

2.2 Model assumptions

- Each coil is an ideal, thin circular filament with uniform current.
- The coils are exactly coaxial, have equal ampere-turns, and are oriented so their axial fields reinforce.
- The magnetic medium is free space, with $\mu_0 = 4\pi \times 10^{-7}\ \mathrm{H\,m^{-1}}$.
- The field map is an x-z plane slice at $y = 0$.
- Working-volume uniformity is sampled on a $9 \times 9 \times 9$ Cartesian grid, with each coordinate between $-50\,\mathrm{mm}$ and $+50\,\mathrm{mm}$.

3 Governing equations

The Biot–Savart differential law for magnetic-field intensity is

$$d\mathbf{H} = \frac{NI}{4\pi}\frac{d\mathbf{l}\times\widehat{\mathbf{R}}}{R^2}, \qquad \mathbf{B} = \mu_0\mathbf{H},$$

where $d\mathbf{l}$ is a directed current-path element and $\mathbf{R}$ points from that element toward the field observation point. Since $\widehat{\mathbf{R}}/R^2 = \mathbf{R}/R^3$, the corresponding differential contribution to flux density is

$$d\mathbf{B} = \frac{\mu_0 NI}{4\pi}\frac{d\mathbf{l}\times\mathbf{R}}{|\mathbf{R}|^3}.$$

For a complete wire path, the total field is the sum (or continuous integral) of these vector contributions. In the numerical model, each coil is represented by a finite set of straight segments, so the integral is approximated by a segment sum.

For a circular coil centered at $z = z_0$, the analytical field on its axis is

$$B_z(z) = \frac{\mu_0 NIa^2}{2\left(a^2 + (z-z_0)^2\right)^{3/2}}.$$

The analytical field of the two-coil pair on the axis is the sum of the two coil contributions:

$$B_{z,\mathrm{pair}}(z) = B_{z,\mathrm{loop}}\left(z; -\frac{d}{2}\right) + B_{z,\mathrm{loop}}\left(z; +\frac{d}{2}\right).$$

For a Helmholtz pair, $d = a$. Substituting $z = 0$ gives the center field

$$B_{\mathrm{center}} = \left(\frac{4}{5}\right)^{3/2}\frac{\mu_0 NI}{a}.$$

This last equation is the basis for the illustrative ampere-turn value.

4 Numerical implementation

4.1 Discretized Biot–Savart calculation

The source code divides each circular coil into N_s equal angular intervals. Each interval is represented by its chord vector $\Delta\mathbf{l}_j$ and midpoint $\mathbf{r}_j$. For an observation point $\mathbf{p}$, the displacement from segment midpoint to observation point is

$$\mathbf{R}_j = \mathbf{p} - \mathbf{r}_j.$$

The approximate field from one coil is then

$$\mathbf{B}(\mathbf{p}) \approx \frac{\mu_0 NI}{4\pi}\sum_{j=1}^{N_s}\frac{\Delta\mathbf{l}_j\times\mathbf{R}_j}{|\mathbf{R}_j|^3}.$$

The pair field is found by calculating both coils and adding their vectors. The numerical function `biot_savart` in `emlab/magnetostatics.py` uses NumPy broadcasting to form the point-to-segment displacement array, computes the cross products and distances, and sums over segments.

Observation points are processed in batches to limit temporary memory use. The batching loop is over groups of field points; the segment contributions themselves are vectorized rather than evaluated in a Python loop.

The implementation returns **B** in tesla. Its `ampere_turns` argument is the product NI for one coil. The analytical helper `on_axis_loop_bz` gives the circular-loop axial field, while `uniformity` calculates the maximum magnitude deviation from a supplied reference field.

4.2 Field uniformity definition

Two different quantities are reported and should not be confused: *numerical discretization error* measures how closely the segment-sum calculation matches an analytical field at the same locations, while *spatial field uniformity* measures how much the magnetic-field magnitude changes from the center to other locations in the working region. For example, the 0.002077303% maximum on-axis numerical-to-analytical error is a discretization comparison; the 0.046914548% on-axis value below is spatial variation of the analytical pair field. A smaller value in one category does not substitute for the other.

For samples with magnitudes B_i and center reference magnitude B_0, the reported maximum relative deviation is

$$U = \frac{\max_i |B_i - B_0|}{B_0} \times 100\%.$$

The on-axis estimate uses the analytical pair field and analytical center field at 41 points from $-50\,\text{mm}$ to $+50\,\text{mm}$. The sampled cube estimate uses the numerical 512-segment-per-coil field at 729 grid points and the numerical field magnitude at the origin as its reference.

5 Analytical and numerical verification

For the illustrative ampere-turn value and $a = 0.35\,\text{m}$, the analytical single-coil field at the center of that coil is

$$B_{\text{single}}(z_0) = \frac{\mu_0 NI}{2a} = 55.901\,699\,437\,\text{mT}.$$

The numerical single-coil center result using 512 segments per coil differs from this analytical value by 0.003137535%. Over the 41 axial samples within the working region, the maximum single-coil numerical-to-analytical axial-field error is also 0.003137535%.

For the Helmholtz pair, the analytical center field is $80.000\,000\,000\,\text{mT}$ by the selected illustrative target. The 512-segment numerical pair calculation gives $80.001\,606\,396\,\text{mT}$, a center error of 0.002007995%. Across the 41 on-axis points from $-50\,\text{mm}$ to $+50\,\text{mm}$, the worst relative difference between the 512-segment numerical axial component and the analytical pair field is 0.002077303%.

5.1 Segment convergence

The segment study compares the numerical pair's B_z against the analytical pair field at the same 41 on-axis locations. The table reports the maximum relative error over those points; all entries are the existing simulation results.

Table 1: Maximum on-axis relative error versus the analytical Helmholtz-pair field for each discretization.

Segments per coil	Maximum relative error (%)
32	0.533333402
64	0.133042648
128	0.033242512
256	0.008309494
512	0.002077303
1024	0.000519321

The error decreases at every refinement. Doubling the number of segments reduces the error by approximately a factor of four in this smooth field region. This is consistent with the second-order behavior expected when approximating a smooth circular path with midpoint/chord segments. It is evidence that the discrete result approaches the analytical result as the wire representation is refined; it does not eliminate the idealized-coil assumptions.

6 Uniformity and symmetry

6.1 On-axis variation: 0.0469%

The maximum on-axis deviation across $z \in [-50\,\text{mm}, 50\,\text{mm}]$, using the analytical pair field and analytical center value as the reference, is 0.046914548%, reported as 0.0469%. In other words, among the sampled axial locations, the largest absolute departure from the center field is about 0.000469 times the center-field magnitude. This small percentage quantifies how flat the pair's axial field is over the specified range; it is not the numerical-versus- analytical discretization error.

6.2 Sampled cube variation: 0.1743%

For the $9 \times 9 \times 9$ grid spanning $\pm 50\,\text{mm}$ in each of x, y, and z, the maximum sampled magnitude deviation from the numerical center-field magnitude is 0.174330570%, or 0.1743%. This is a three-dimensional sampled-cube metric, not an on-axis metric. The cube includes off-axis points, where the field can vary more than on the symmetry axis. Since it uses a finite grid, it estimates the largest deviation among those 729 points and is not a proof of the continuous-volume maximum.

6.3 Transverse components on the symmetry axis

For an ideal coaxial circular coil, rotational symmetry about the z-axis leaves no preferred transverse direction at an axial observation point. Contributions to B_x and B_y from symmetrically placed current elements cancel, while the axial contributions add. The same symmetry applies to the coaxial pair, so ideally $B_x = B_y = 0$ on the axis. The simulation's maximum transverse-component magnitude is $1.212296943 \times 10^{-17}$ T, effectively zero at the scale of the modeled field. The tiny residual is consistent with finite angular discretization and floating-point roundoff.

7 Figures

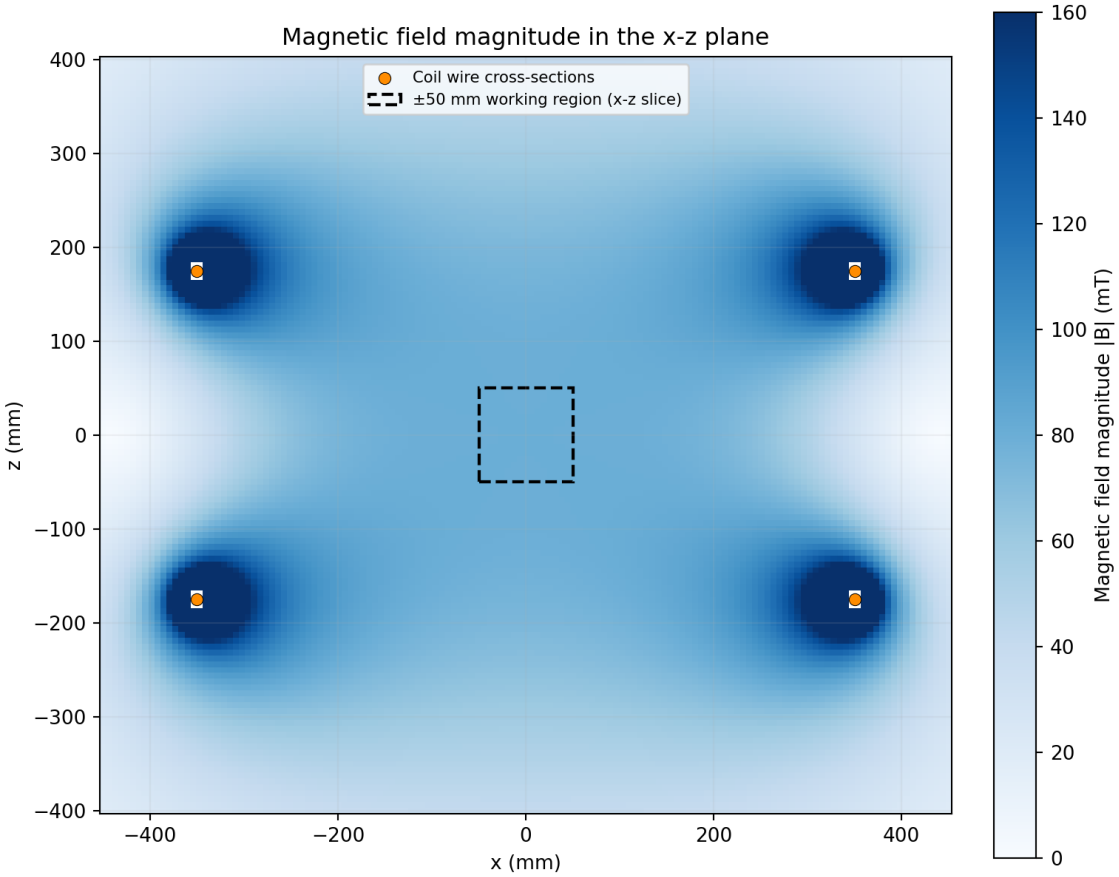


Figure 1: Magnetic-field magnitude $|\mathbf{B}|$ in the x-z plane ($y = 0$), in millitesla, for the illustrative field case. Orange markers show the coil-wire cross-sections in the slice. The dashed square marks the ± 50 mm working-region cross-section. The ideal filament's immediate neighborhood is masked because its field is singular on the wire.

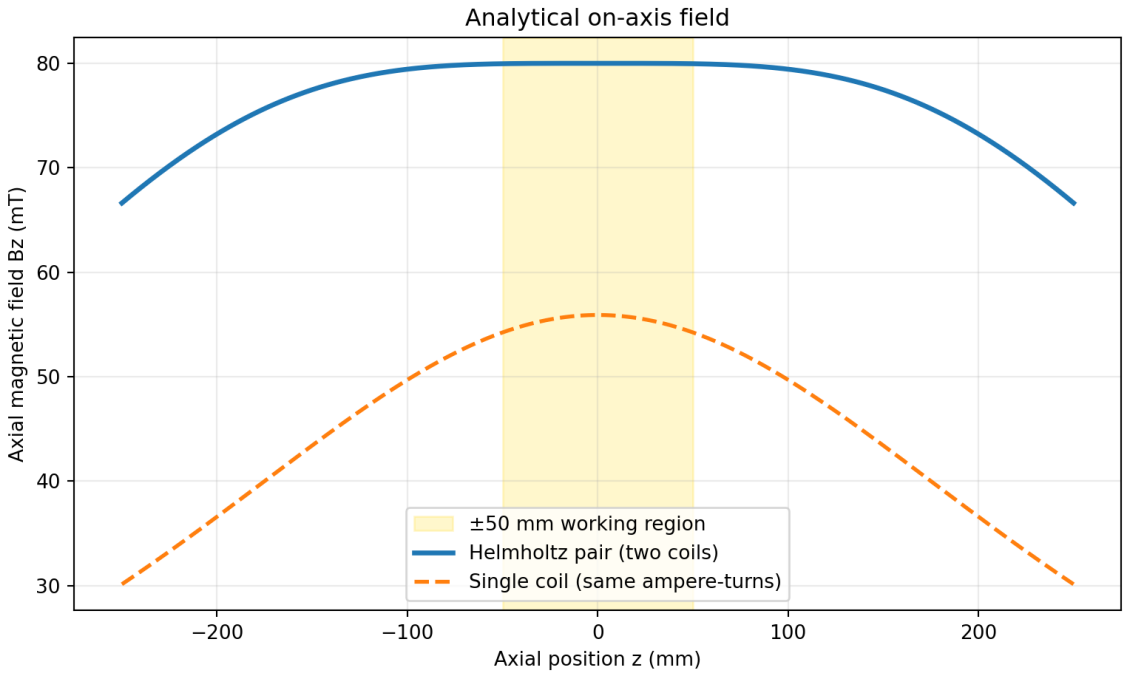


Figure 2: On-axis comparison between the analytical Helmholtz-pair field and one circular coil with the same ampere-turns as either member coil. The horizontal axis spans approximately $-250\,\mathrm{mm}$ to $+250\,\mathrm{mm}$; the highlighted central interval is the $\pm 50\,\mathrm{mm}$ working region. The pair's central profile is flatter than the single-coil profile.

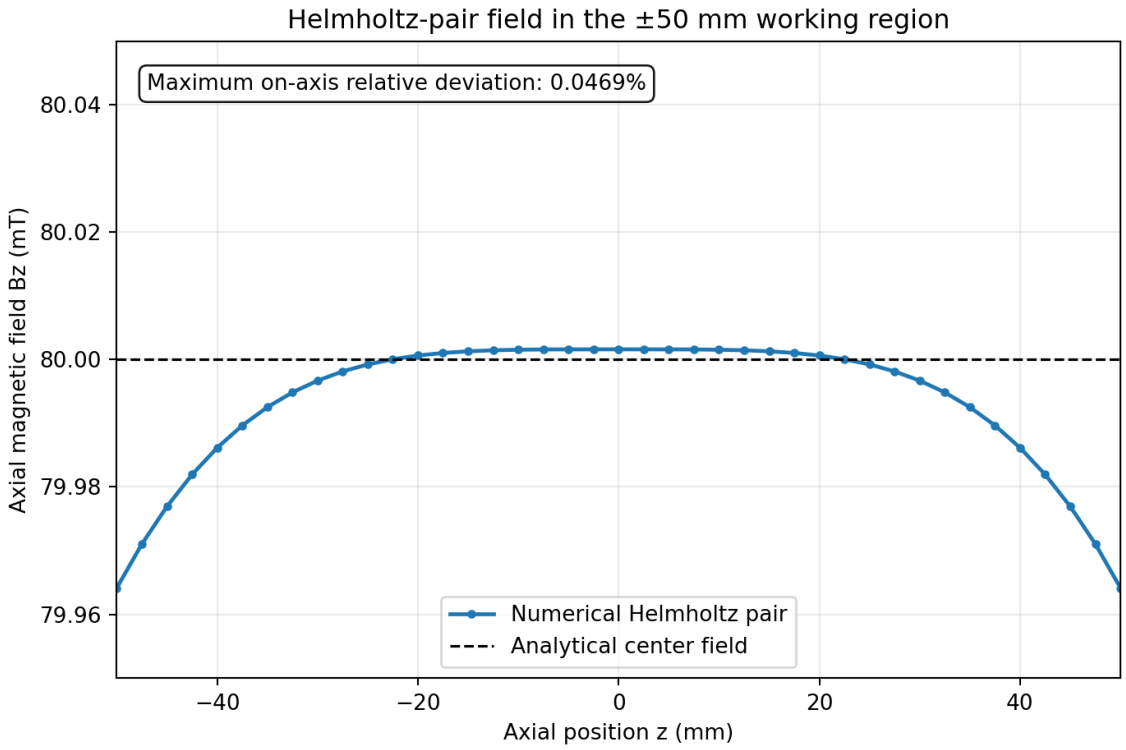


Figure 3: Zoomed Helmholtz-pair axial field across the ±50 mm working interval. Markers show the 41 existing numerical Biot–Savart samples; the dashed line is the analytical center field. The annotation reports the calculated 0.0469% maximum on-axis variation. The vertical scale is intentionally restricted to 79.95 mT through 80.05 mT to make the small variation visible.

8 Convergence, assumptions, and limitations

The convergence table demonstrates decreasing numerical error as the circular path is resolved with more segments. The reported accuracy concerns the discretized Biot–Savart sum compared with the analytical field at selected points. It should not be confused with the *field uniformity* percentages, which describe how the field varies spatially within a working region.

The model idealizes each coil as a zero-thickness filament and omits finite wire diameter, winding thickness, connecting leads, nearby magnetic materials, alignment tolerances, and current-source limits. It assumes free-space permeability and perfectly equal coil currents. The field-map values near the wires are especially sensitive to the filament idealization; the plotted map masks a small region around the wire locations for readability and to avoid displaying the ideal singularity.

The numerical calculations and figures document an illustrative setup only. The 80 mT case and inferred NI must not be represented as official project values until compared with the professor's starter kit. No professor-provided tests or official starter files are represented as having been run or verified in this report. In particular, this report makes no claim that professor acceptance tests pass; those tests were not available and have not been executed.

9 How I Would Explain This Project to My Professor

I modeled each Helmholtz coil as a circular wire and divided it into many short segments. At each field point, the code uses the Biot–Savart law to calculate the vector contribution from every segment, sums those contributions for each coil, and adds the two coil fields. I then checked the result against the analytical field on a circular coil's axis and the analytical Helmholtz center-field formula.

The numerical error decreased from about 0.5333% with 32 segments per coil to 0.0005193% with 1024 segments per coil, which shows that the discrete calculation converges toward the analytical answer as the coil representation is refined. The pair's on-axis field varies by only 0.0469% over ±50 mm; over the sampled three-dimensional cube, the maximum sampled variation is 0.1743%. Symmetry predicts zero transverse field on the axis, and the very small numerical remainder is consistent with discretization and roundoff. I would also clarify that the 80 mT input is illustrative until the professor's official current and turn values are available.

10 Reproducibility

The source files used for this work include the original `m1_standalone.py`, the package implementation `emlab/magnetostatics.py`, the explanatory `report/milestone1_notes.md`, and `report/milestone1_run_results.txt`. The three figures are included from the project's `figures/` directory. The organized Milestone 1 script can be run from the project root using:

```
source .venv/bin/activate
python -m scripts.m1_navigation_field
```

This regenerates the graphs and numerical results using the current project inputs. It does not validate those inputs against the missing official starter kit.

11 Note on the final course report

This Milestone 1 document is intentionally maintained as detailed working documentation and is not the final consolidated course report. The final course report must bring together results from all five milestones and meet the professor's required total length of 8–12 pages. When preparing that submission, this material should be edited and integrated with the other milestones to fit the required page limit without treating this longer Milestone 1 record as the final deliverable.